

Figure 1: Program logic

```
dict1[file]= root
```

The *create* function opens the thread process for each drive, and each thread process calls the *search1* function.

```
def create():
    t1= datetime.now()
    list2 = [] # empty list is created
    list1 = get_drives()
    print list1
    for each in list1:
        process1 = Thread(target=search1, args=(each,))
        process1.start()
        list2.append(process1)
```

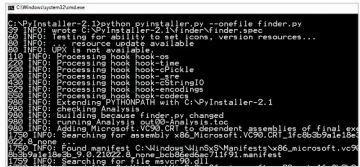
```
for t in list2:
    t.join() # Terminate the threads
```

After creating the dictionary, the following code dumps the dictionary into the hard disk as a *pickle* file.

```
pickle_file = open("finder_data","w")
cPickle.dump(dict1,pickle_file)
pickle_file.close()
t2= datetime.now()
total =t2-t1
print "Time taken to create " , total
print "Thanks for using L4wisdom.com"
```

Next time, when you search any file, the program will search the file in the dumped dictionary, as follows:

```
if len(sys.argv) < 2 or len(sys.argv) > 2:
    print "Please use proper format"
    print "Use <finder -c > to create database file"
    print "Use <finder file-name> to search file"
    print "Thanks for using L4wisdom.com"
```

Figure 2: Creating *exe* file of the Python program

```
elif sys.argv[1] == '-c':
    create()
else:
    t1= datetime.now()
    try:
        pickle_file = open("finder_data", "r")
        file_dict = cPickle.load(pickle_file)
        pickle_file.close()
    except IOError:
        create()
    except Exception as e:
        print e
        sys.exit()
    file_to_be_searched = sys.argv[1].lower()
    list1= []
    print "Path \\t\\t: File-name"
```

Here, we used the search method of regular expressions so that we can use a regular expression to find the file.

```
for key in file_dict:
    if re.search(file_to_be_searched, key):
        str1 = file_dict[key]+ " : "+key
        list1.append(str1)
list1.sort()
for each in list1:
    print each
    print "-----"
t2= datetime.now()
total =t2-t1
print "Total files are", len(list1)
print "Time taken to search " , total
print "Thanks for using L4wisdom.com"
```

The rest of the code is very easy to understand.

Let us save the complete code as *finder.py* (you can also download it from http://opensourceforu.com/article_source_code/sept16/finder.zip) and make it a Windows executable (*exe*) file using the *Pyinstaller* module. You can also download it from http://l4wisdom.com/finder_go.php. Run the command shown in Figure 2. After running it successfully, you can find the *finder.exe* in folder *C:\PyInstaller-2.1\finderdist*.